WESTERN FRONT

Paper 1: Medicine Through Time with the Western Front

Name: ______________________ Class: ______________

PROGRESS & ASSESSMENT TRACKER

Lesson / Assessment Title	Effort	Level	Teacher Comments
L1: KT5.1: What was the historical and medical context of the Western Front?			
L2: KT5.2: How did the trench environment create new medical challenges?			
L3: KT5.3: What new illnesses and wounds did soldiers face?			
L4: KT5.4: How did the RAMC and FANY operate the chain of evacuation?			
L5: KT5.5: What incredible medical advances were forged on the Western Front?			
Final Unit Grade:			

L1: KT5.1: What was the historical and medical context of the Western Front?

(See Textbook Page 3)

LEARNING OBJECTIVES

- [ ] Explain the state of surgical asepsis, radiology, and transfusion science on the eve of World War I.

Do Now Activity

Score: / 10

1. What lifestyle choice was definitively linked to lung cancer in the 1950s?

2. Name one modern technology used to diagnose lung cancer.

3. State one way the modern government tries to prevent people from smoking.

4. When did the First World War begin and end?

5. What was the 'Western Front'?

6. Name the three major battles fought by the British on the Western Front.

7. What disease did Edward Jenner vaccine against in 1796?

8. What was established in Britain in 1948 to provide free medical care?

9. Why did the terrain of Ypres make evacuation and medical treatment very difficult?

10. What did the battle of Cambrai (1917) demonstrate about tanks?

Vocabulary Check

Fill in the blanks using the vocabulary words below.

Aseptic surgery | Mobile X-ray Unit | Blood Depot

Before the outbreak of the First World War, major breakthroughs in Britain laid the foundation for wartime medical advances. In operating theatres, surgeons transitioned to ______________ to prevent microbes from entering wounds during procedures. Meanwhile, the discovery of radiation allowed for the use of ______________ designs to find deeply embedded shrapnel in wounded soldiers closer to the battlefield. Finally, to treat severe blood loss and shock, doctors developed methods of preserving and storing blood in a ______________ so that transfusions could be carried out quickly without the donor needing to be present.

Q1. Identify the main geographical feature of the Western Front.

Q2. Describe the conditions of the trenches during the war of attrition.

Q3. Explain how artillery bombardment worsened the medical situation.

Q4. To what extent was the RAMC prepared for the medical challenges of the Western Front?

KNOWLEDGE CHECK (Q5)

What was the main limitation of performing blood transfusions in hospital settings before the discovery of sodium citrate in 1915?

- [ ] A. Blood types had not yet been discovered, so transfusions were always fatal.

☐ B. Blood could not be stored because it quickly coagulated (clotted), meaning a donor had to be physically present.

☐ C. Syringes had not yet been invented.

☐ D. The Church banned the transfer of blood between two people.

Pair & Share Activity

Q6. Prompt: Which pre-war medical development was the most critical for the survival of soldiers on the Western Front: the rise of aseptic surgery, the invention of mobile X-rays, or the progress in storing and transfusing blood?

Think: Jot down your choice and one piece of evidence from your knowledge (such as the high rate of infection from muddy trench soil making aseptic methods vital, or blood transfusions preventing fatal shock before surgery).

Your Notes:

Historian's Corner: The Debate over the Source of Medical Innovation in World War I

Historians of military medicine debate whether the rapid medical advances of the First World War represented brand new, revolutionary breakthroughs or merely the accelerated application of existing peacetime technologies. Traditionalist historians argue that the Western Front acted as a vast clinical laboratory, forcing rapid, unprecedented experimentation that yielded genuine breakthroughs in surgery and logistics. However, revisionist historians argue that almost all key technologies-including aseptic surgery, sodium citrate blood preservation, and portable X-ray tubes-had already been discovered in civilian laboratories before 1914. They suggest that the war's real contribution was not scientific creation, but rather the massive state funding and logistical mobilization that allowed these pre-existing civilian ideas to be scaled up and standardized across thousands of casualties.

GCSE Exam Practice

Source A

From the personal pocket diary of Private Thomas Atkins, serving with the British Infantry in the Ypres Salient, November 1916.

"The communication trenches are flooded with three feet of icy water, and the clay walls are collapsing into thick slime. Our boots are permanently waterlogged, and we have to dig out our firesteps constantly. The enemy shelled our support lines yesterday, throwing up colossal geysers of wet mud that choked our drainage channels."

Source B

An official archival photograph taken on the Somme in 1916, showing British infantrymen in a muddy communication trench.

Provenance Scaffolding:

Source A is a private diary from a soldier on the front line; consider how this affects its honesty compared to an official account. Source B is an official photograph, which might be censored, but what does the undeniable presence of thigh-deep mud tell you regardless?

Exam Q46. 1 (a). How useful are Sources A and B for an enquiry into the difficulties faced by soldiers in the trench system of the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)

Exam Q47. Describe one feature of the First Battle of Ypres. (2 marks)

Exam Q48. Describe one feature of the Battle of the Somme. (2 marks)

> **Exam Q49. Describe one feature of the support trench system on the Western Front. (2 marks)**

Exam Q50. Describe one feature of the problems involved in transporting wounded soldiers away from the battleground. (2 marks)

Exam Q51. How useful are Sources A and B for an enquiry into the difficulties faced by soldiers in the trench system of the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)

Source A

Source A: From the personal pocket diary of Private Thomas Atkins, serving with the British Infantry in the Ypres Salient, November 1916. 'The communication trenches are flooded with three feet of icy water, and the clay walls are collapsing into thick slime. Our boots are permanently waterlogged, and we have to dig out our firesteps constantly. The enemy shelled our support lines yesterday, throwing up colossal geysers of wet mud that choked our drainage channels.'

Source B

Source B: An official photograph taken on the Somme in October 1916. It shows British soldiers attempting to maintain a communication trench.

Exam Q52. How could you follow up Source A to find out more about the work of the stretcher bearers on the Western Front? (4 marks)

Source A

Source A: A diary entry from an RAMC officer at the Somme, 1916: 'The terrain is devastated. We cannot use the motor ambulances as the heavy rains and artillery have turned the roads into deep quagmires of mud. We are forced to use six horses to pull a single wagon.'

General Notes

L2: KT5.2: How did the trench environment create new medical challenges?

(See Textbook Page 7)

LEARNING OBJECTIVES

- ☐ Explain the causes, symptoms, and preventive strategies developed to combat trench foot and trench fever.
- ☐ Analyze the impact of weaponry on injuries, specifically focusing on shrapnel, head injuries, and gas gangrene from soil.

Do Now Activity

Score: / 10

1. When did the First World War begin and end?

2. What was the 'Western Front'?

3. Name the three major battles fought by the British on the Western Front.

4. What was 'No Man's Land'?

5. Name the pattern in which trenches were dug.

6. What was the purpose of the 'duckboards' in the trenches?

7. Who discovered the cause of cholera in 1854?

8. Which anesthetic was discovered by James Simpson in 1847?

9. Explain why the zig-zag pattern of trenches was crucial for survival.

10. What were the three main parallel lines of trenches?

Vocabulary Check

Write a historically accurate sentence connecting two terms from the glossary box below.

Trench Foot | Trench Fever | Shell Shock

Your Sentence:

Q1. Identify the cause of Trench Foot.

Q2. Describe the symptoms of Trench Fever.

Q3. Explain how the army attempted to prevent Trench Fever.

Q4. Evaluate the effectiveness of military discipline in preventing environmental illnesses.

KNOWLEDGE CHECK (Q5)

How did the British Army attempt to prevent soldiers from developing trench foot in wet and muddy conditions?

☐ A. By providing them with waterproof rubber wellington boots.

- [ ] B. By issuing orders for soldiers to rub their feet with whale oil and regularly change their socks.
- [ ] C. By administering a trench foot vaccine to all soldiers.
- [ ] D. By heating the trenches with underground pipes.

Pair & Share Activity

Q6. Prompt: Which aspect of the trench environment presented the most difficult challenge for the Royal Army Medical Corps (RAMC): the physical environment (producing trench foot), the biological vectors (producing trench fever), or the psychological strain of industrial warfare (producing shell shock)?

Think: Jot down your choice and one piece of evidence from the sources, such as the thousands of men evacuated for trench foot, the difficulty of delousing crowded trenches to stop lice, or the fact that shell shock was a completely new condition requiring specialists.

Your Notes:

Historian's Corner: The Debate over the Treatment and Diagnosis of Shell Shock (NYD.N)

Historians of military medicine actively debate the British Army's evolving response to shell shock (NYD.N) during the First World War. Traditionalist histories often focused on the harshness of the military command, highlighting that early in the war, soldiers showing symptoms of psychological trauma were frequently accused of cowardice or malingering, with some even facing court-martial and execution. However, revisionist historians argue that the British medical services adapted remarkably quickly to an unprecedented crisis. They point out that by 1916, the RAMC recognized shell shock as a genuine medical condition, establishing specialist treatment centers close to the front lines. They argue that the policy of treating soldiers quickly and locally (focusing on rest and reassurance) was a highly advanced clinical approach that aimed to prevent chronic mental illness, even though the primary military motive remained returning men to active fighting.

GCSE Exam Practice

Source A

From the personal memories of William Easton, an 18-year-old volunteer with the East Anglian Field Ambulance, describing the Passchendaele/Ypres campaign in 1917.

"During the third offensive at Ypres, we navigated paths choked with thick, hip-deep mire. In the devastated sector of Hooge, we struggled to relocate casualties. The physical exertion of a single carry left us utterly spent; delivering just two or three wounded men to safety daily was our absolute limit. We worked in teams of four, using shoulder straps to stabilize the stretcher and prevent agonising falls."

Source B

An official archival photograph taken on the Somme in 1916, showing a horse-drawn ambulance navigating a deeply rutted, waterlogged dirt road.

Provenance Scaffolding:

Think about who wrote or produced this source. For Source A, it's an East Anglian Field Ambulance worker at Ypres. Does this personal, on-the-ground experience make it more reliable for understanding the physical toll? For Source B, it's an official photograph from the Somme. Was it meant to document reality, or perhaps carefully framed for morale?

Exam Q53. 2 (a). How useful are Sources A and B for an enquiry into the impact of the terrain on the transport of the wounded on the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)

Exam Q54. Describe one feature of the trench system. (2 marks)

Exam Q55. Describe one feature of trench foot. (2 marks)

Exam Q56. Describe one feature of poison gas attacks on the Western Front. (2 marks)

Exam Q57. Describe one feature of the fighting on the Western Front that led to a high number of injuries. (2 marks)

Exam Q58. How useful are Sources A and B for an enquiry into the problem of ill health arising from the trench environment? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)

Source A

Source A: A report from a Medical Officer in 1915: 'Altogether about 200 men had to be evacuated from our section of the trenches with swollen, numb feet due to the freezing mud. Several require amputation.'

Source B

Source B: A diary entry from an RAMC surgeon: 'The men are brought in with horrific shrapnel injuries. The metal drives the filthy, manured soil deep into the muscle, creating an environment where gas gangrene thrives. We often have to amputate to save their lives.'

Exam Q59. How could you follow up Source A to find out more about the problem of trench foot? (4 marks)

Source A

Source A: A report from a Medical Officer in 1915: 'Altogether about 200 men had to be evacuated from our section of the trenches with swollen, numb feet due to the freezing mud. Several require amputation.'

General Notes

L3: KT5.3: What new illnesses and wounds did soldiers face?

(See Textbook Page 11)

LEARNING OBJECTIVES

- [ ] Explain the connection between artillery weaponry, shrapnel, and the introduction of the steel Brodie helmet in 1915.
- [ ] Analyse how the muddy, manured environment of Flanders led to severe wound infections like gas gangrene and tetanus.

Do Now Activity

Score: / 10

1. What was 'No Man's Land'?

2. Name the pattern in which trenches were dug.

3. What was the purpose of the 'duckboards' in the trenches?

4. What is 'Trench Foot'?

5. Which weapon caused the most casualties on the Western Front?

6. Why were wounds on the Western Front highly prone to severe infection?

7. Who discovered Penicillin by accident in 1928?

8. What structure did Watson and Crick discover in 1953?

9. What preventative measure was introduced to stop Trench Foot?

10. What was 'Shell Shock'?

Vocabulary Check

Write a clear definition and a historically accurate sentence for the term: ***Gas Gangrene***

Q1. Identify the weapon that caused over 60% of all wounds on the Western Front.

Q2. Describe the physical and psychological effects of poison gas.

Q3. Explain why shrapnel wounds almost always became infected.

KNOWLEDGE CHECK (Q4)

Why were deep shrapnel wounds on the Western Front almost guaranteed to develop the deadly infection known as gas gangrene?

- ☐ A. Because the German army dipped their bullets in poison before firing them.
- ☐ B. Because the French farmland had been heavily fertilized with manure for centuries, filling the soil with deadly tetanus and gangrene bacteria.
- ☐ C. Because the British soldiers refused to wash their uniforms.
- ☐ D. Because chlorine gas from chemical attacks infected the wounds.

Q5. To what extent could doctors successfully treat deep infections during WWI?

Pair & Share Activity

Q6. Prompt: Which posed a more difficult medical challenge for the Royal Army Medical Corps (RAMC) on the Western Front: the physical injuries caused by industrial weaponry (such as shrapnel wounds and head trauma) or the environmental infections caused by the Flanders terrain (such as gas gangrene, tetanus, and trench foot)?

Think: Jot down your choice and one piece of evidence from the sources, such as the introduction of the Brodie helmet reducing head wounds vs. the manured soil causing gas gangrene infections that standard antiseptics couldn't treat.

Your Notes:

Historian's Corner: The Debate over the Diagnosis and Treatment of Shell Shock (NYD.N)

Historians of military medicine are deeply divided over the British Army's response to psychological trauma, which was medically recorded as shell shock or NYD.N ('Not Yet Diagnosed, Nervous'). Traditionalist historians argue that the military high command was unsympathetic and hostile, often viewing traumatized soldiers as cowards or malingerers, which in some instances led to court-martials and executions for desertion. Conversely, revisionist historians argue that the RAMC adapted with remarkable speed to an entirely new clinical phenomenon. They highlight that by 1916, the army had established specialized psychiatric units close to the front line, utilizing the advanced concept of 'proximity, immediacy, and expectation' (treating men locally so they expected to return to their units), which laid the groundwork for modern trauma therapy.

GCSE Exam Practice

Source A

From a wartime notebook of Lance Sergeant Elmer Cotton in 1915, providing a private, firsthand account of a gas attack.

"A chlorine discharge floods the victim's respiratory tract, effectively suffocating them while standing on dry ground. The resulting symptoms are excruciating: a blinding headache, an overwhelming craving for water-though drinking it brings immediate fatality-and a severe, stabbing sensation in the lungs. Patients continuously expectorate a distinctive greenish fluid. It represents a truly horrific way to perish."

Source B

An official photograph capturing blinded soldiers forming a chain outside an Advanced Dressing Station.

Provenance Scaffolding:

For Source A, consider that it is from a wartime notebook of Lance Sergeant Elmer Cotton in 1915. This is a private, unedited firsthand account. How does this affect its usefulness? For Source B, an official photograph showing blinded soldiers forming a chain. Photographs capture reality but can also be staged. Does the clear visual evidence of the gas's impact make it highly useful regardless?

Exam Q60. 2 (a). How useful are Sources A and B for an enquiry into the effects of poison gas attacks on soldiers on the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)

Exam Q61. Describe one feature of the effects of poison gas. (2 marks)

Exam Q62. Describe one feature of shrapnel wounds. (2 marks)

Exam Q63. Describe one feature of aseptic surgery in the early 20th century. (2 marks)

Exam Q64. Describe one feature of early x-ray machines. (2 marks)

Exam Q65. How could you follow up Source A to find out more about the treatment and prevention of the effects of gas attacks? (4 marks)

Source A

Source A: From an official report written by a medical officer at an Advanced Dressing Station near Ypres, May 1915: 'At 4:00 AM, the enemy released a dense cloud of yellowish-green gas toward our trenches. Many of the men did not adjust their woolen flannel pads quickly enough. Over fifty casualties were brought into our Dressing Station, their eyes swollen shut and blinded by the vapor. We washed their eyes with bicarbonate solution, but several have already suffocated.'

General Notes

L4: KT5.4: How did the RAMC and FANY operate the chain of evacuation?

(See Textbook Page 15)

LEARNING OBJECTIVES

- ☐ Explain the four main stages of the Western Front chain of evacuation and the mode of transport between them.
- ☐ Evaluate the roles, significance, and contributions of the RAMC and FANY on the Western Front.

Do Now Activity

Score: / 10

1. What is 'Trench Foot'?

2. Which weapon caused the most casualties on the Western Front?

3. Why were wounds on the Western Front highly prone to severe infection?

4. What does RAMC stand for?

5. What was the role of the FANY?

6. Describe the 'Chain of Evacuation' on the Western Front.

7. What lifestyle choice was definitively linked to lung cancer in the 1950s?

8. What was established in Britain in 1948 to provide free medical care?

9. What was the primary role of the Casualty Clearing Station (CCS)?

10. Where were the Base Hospitals located and what did they do?

Vocabulary Check

Fill in the blanks using the vocabulary words below.

Chain of Evacuation | Casualty Clearing Station (CCS) | First Aid Nursing Yeomanry (FANY)

To save soldiers wounded in the trenches, they had to be moved quickly along the ________________ before soil-borne bacteria caused fatal infections. While basic first aid was given close to the battlefield, soldiers with life-threatening wounds were sent further back to a ________________ for urgent surgery. To transport these casualties safely, the army relied on volunteers from the ________________ , who braved shellfire to drive motorized ambulances between frontline dressing stations and the rear hospitals.

Q1. Identify the first stage of the Chain of Evacuation.

Q2. Describe the role of Casualty Clearing Stations (CCS).

Q3. Explain how patients were transported between the different stages of the evacuation chain.

Q4. Evaluate the significance of the Chain of Evacuation.

KNOWLEDGE CHECK (Q5)

What was the main difference between the medical treatments performed at a Regimental Aid Post (RAP) and a Casualty Clearing Station (CCS)?

☐ A. The RAP performed complex surgery, while the CCS only applied bandages.

☐ B. The RAP provided immediate first aid near the frontline, while the CCS was equipped with operating theatres and X-ray machines to perform life-saving surgery.

☐ C. The RAP was for officers only, while the CCS was for regular soldiers.

☐ D. The RAP was located in Britain, while the CCS was in France.

Pair & Share Activity

Q6. Prompt: Was the medical service's success on the Western Front driven more by the highly organized military logistics of the RAMC, or by the courage and adaptability of voluntary organizations like the FANY?

Think: Jot down your choice and one piece of evidence from your knowledge (such as the systematic stages of triage from RAP to Base Hospital vs. FANY volunteers driving motorized ambulances under active bombardment).

Your Notes:

Historian's Corner: The Debate over the Primary Role of Casualty Clearing Stations

Historians of military medicine actively debate how the roles of Casualty Clearing Stations (CCSs) and Base Hospitals shifted during the First World War. Traditional accounts argued that Base Hospitals, situated near the safe French coast, remained the primary hubs for surgical intervention throughout the war. However, revisionist historians have shown that the extreme threat of gas gangrene and tetanus from Flanders soil forced a rapid role reversal. Because debridement (wound excision) had to be performed within hours of injury, CCSs located near the frontline railheads became the primary hubs for life-saving surgery, while Base Hospitals were relegated to long-term convalescence. This adaptation demonstrates that wartime medical logistics was not a static plan but a dynamic, rapidly evolving system.

GCSE Exam Practice

Source A

From the 1919 published autobiography of Pat Beauchamp, a volunteer FANY ambulance driver.

"With the Somme offensive raging, dozens of transport cars would form a convoy to meet incoming hospital trains at the railway siding. Once positioned, stretcher bearers loaded our vehicles with four severely injured, non-walking cases. Traversing the ruined, potholed cobblestone roads was an agonizing ordeal. The violent jolts caused immense suffering, yet we were forced to drive slowly to protect those inside."

Source B

A 1916 archival photograph from the Somme showing a motor ambulance navigating a flooded road.

Provenance Scaffolding:

Source A is from the 1919 published autobiography of a FANY ambulance driver. It's a memoir published right after the war. How might writing for a public audience affect her description of the difficulties? Source B is a 1916 photograph from the Somme. Official photographs were subject to censorship. Does its depiction of flooded roads match your own knowledge of the Somme battlefield?

Exam Q66. 2 (a). How useful are Sources A and B for an enquiry into the problems of transporting the wounded on the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)

Exam Q67. Describe one feature of the FANY. (2 marks)

Exam Q68. Describe one feature of the Chain of Evacuation. (2 marks)

Exam Q69. Describe one feature of the underground hospital at Arras. (2 marks)

Exam Q70. Describe one feature of the work of the FANY on the Western Front. (2 marks)

Exam Q71. How useful are Sources A and B for an enquiry into the work of medical staff in the Casualty Clearing Stations (CCS) on the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)

Source A

Source A: A letter from an RAMC nurse working at a Casualty Clearing Station (CCS) in 1916: 'We worked night after night, in the thunderous noise of raging battles. The surgeons are constantly amputating limbs to stop the spread of gangrene before putting the men on the trains.'

Source B

Source B: A diary entry from a member of the FANY in 1917: 'I drove my motor ambulance through the terrible mud for 14 hours straight today, bringing men from the Main Dressing Station down to the railhead.'

Exam Q72. How could you follow up Source A to find out more about the work of medical staff in the Casualty Clearing Stations? (4 marks)

Source A

Source A: A letter from an RAMC nurse working at a Casualty Clearing Station (CCS) in 1916: 'We worked night after night, in the thunderous noise of raging battles. The surgeons are constantly amputating limbs to stop the spread of gangrene before putting the men on the trains.'

General Notes

L5: KT5.5: What incredible medical advances were forged on the Western Front?

(See Textbook Page 19)

LEARNING OBJECTIVES

- [ ] Explain how the Thomas splint and mobile X-ray units solved critical evacuation and surgical challenges.
- [ ] Analyze the development of blood transfusion and the significance of the first blood bank at the Battle of Cambrai.

Do Now Activity

Score: / 10

1. What does RAMC stand for?

2. What was the role of the FANY?

3. Describe the 'Chain of Evacuation' on the Western Front.

4. What was the Thomas Splint?

5. What chemical was discovered in 1915 to prevent blood from clotting?

6. What did the discovery of blood storage allow surgeons to do?

7. Who discovered the structure of DNA in 1953?

8. Who introduced antiseptic surgery using carbolic acid in 1865?

9. How did WWI affect the use of X-rays in medicine?

10. What was the 'Carrel-Dakin' method?

Vocabulary Check

Write a historically accurate sentence connecting two terms from the glossary box below.

Thomas Splint | Carrel-Dakin Method | Blood Depot

Your Sentence:

Q1. Identify the device designed by Hugh Owen Thomas.

Q2. Describe how Richard Lewisohn improved blood transfusions.

KNOWLEDGE CHECK (Q3)

How did the implementation of the Thomas splint on the Western Front in 1916 directly reduce mortality rates among wounded soldiers?

- [ ] A. It stopped the spread of infection by killing the bacteria that caused gas gangrene.
- [ ] B. It cured shell shock by holding the soldier firmly in place.
- [ ] C. It kept fractured leg bones rigidly in place during transport, preventing fatal blood loss and shock.
- [ ] D. It provided a steady drip of blood for transfusions on the battlefield.

Q4. Explain how the Thomas Splint improved survival rates.

Q5. To what extent did the Western Front accelerate medical progress?

Pair & Share Activity

Q6. Prompt: Which medical advance on the Western Front represented a more impressive leap forward: the mechanical simplicity of the Thomas Splint which stabilized fractures, or the complex chemical engineering of blood preservation and storage?

Think: Jot down your choice and one piece of evidence from the sources (e.g., the Thomas splint keeping the leg rigid to prevent internal bleeding during transport, or sodium citrate allowing blood to be stored in depots for the first time without donors present).

Your Notes:

Historian's Corner: The Debate over the Western Front as a Catalyst for Medical Innovation

Military and social historians actively debate the true significance of the First World War in driving medical progress. Traditional histories argue that the Western Front served as a massive, unprecedented laboratory of rapid innovation, where the extreme pressures of trench warfare forced doctors to experiment and make revolutionary breakthroughs in antiseptics, radiology, and blood storage. Conversely, revisionist historians argue that almost all of these 'wartime breakthroughs' were actually based on pre-war civilian discoveries that had already been established in laboratories before 1914, such as the basic design of the Thomas splint, Marie Curie's work with radioactivity, and pre-war transfusion experiments. From this perspective, the war did not spark scientific creation itself, but rather acted as an institutional and financial accelerator, providing the state funding and millions of casualties necessary to scale up and standardize pre-existing civilian technologies.

GCSE Exam Practice

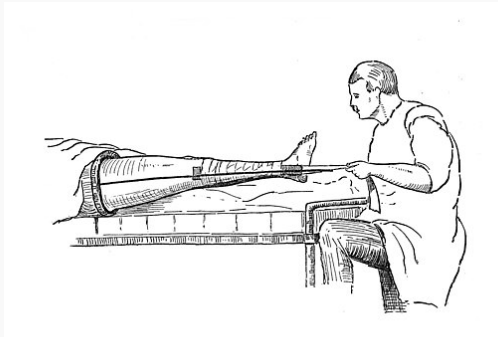
Source A

From an official clinical report written by Captain Arthur Vaughan, a surgeon serving with the Royal Army Medical Corps (RAMC) at a Casualty Clearing Station (CCS) behind the frontline, October 1916.

"Our medical team at the Casualty Clearing Station has fully adopted the new leg traction splint. Before we received this iron frame, almost every compound femur fracture arrived at our ward in a state of terminal shock due to the jagged bones grating inside the flesh during transit. Since implementing this immobilization frame last month, we have kept the fractured limbs completely rigid. Out of forty admissions with thigh wounds, we have lost only six patients."

Source B

A studio photograph taken to train medical orderlies, showing a soldier's leg secured in a Thomas splint.



Provenance Scaffolding:

Source A is a clinical report by a surgeon; does this make its statistics reliable? Source B is a staged training photograph; while not taken on the chaotic frontline, does this actually make it better for demonstrating how the device technically worked?

Exam Q73. 2 (a). How useful are Sources A and B for an enquiry into the effectiveness of new medical developments used to treat wounded soldiers on the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)

Exam Q74. Describe one feature of the Thomas Splint. (2 marks)

Exam Q75. Describe one feature of the medical advances in blood transfusions on the Western Front. (2 marks)

Exam Q76. How useful are Sources A and B for an enquiry into the effectiveness of new medical developments used to treat wounded soldiers on the Western Front? Explain your answer, using Sources A and B and your knowledge of the historical context. (8 marks)

Source A

Source A: From an official clinical report written by Captain Arthur Vaughan, a surgeon serving with the Royal Army Medical Corps (RAMC) at a Casualty Clearing Station (CCS) behind the frontline, October 1916.
'Our medical team at the Casualty Clearing Station has fully adopted the new leg traction splint. Before we received this iron frame, almost every compound femur fracture arrived at our ward in a state of terminal shock due to the jagged bones grating inside the flesh during transit. Since implementing this immobilization frame last month, we have kept the fractured limbs completely rigid. Out of forty admissions with thigh wounds, we have lost only six patients.'

Source B

Source B: A photograph from a British medical instruction manual, published in 1917, showing the deployment of a traction device.
(Visual Description: A detailed, black-and-white studio photograph showing a soldier lying flat on a wooden stretcher. His right leg is suspended and held rigid inside a metallic Thomas splint, which consists of a padded leather ring pushed up against the groin and parallel steel rods running down both sides of the limb. Bandages and canvas straps are wrapped tightly around the thigh and calf to keep the leg completely straight. A canvas sling is hooked to the end of the splint to pull the foot forward, creating traction. Two medical orderlies in clean aprons stand alongside, demonstrating how to attach the secure traction cords.)

Exam Q77. How could you follow up Source A to find out more about new techniques being used on the Western Front to deal with injuries? (4 marks)

Source A

Source A: A report from a surgeon at a CCS in 1916: 'Several unsuccessful trials this morning to extract a shell fragment from a soldier's brain. Finally, we decided to try using a large wire nail as a magnet.'

General Notes

Factors Overview: Western Front

Edexcel focuses heavily on the factors that drove medical progress (or held it back). For each factor below, write one specific historical example from this period that either helped or hindered medical progress.

Factor	Specific Historical Example & Impact
Individuals	
The Church & Religion	
Government & Wealth	
Science & Technology	
Attitudes in Society	
War	